COVID-19 Visual Analytics Using Tableau: A Multi-Dashboard Study of India's Epidemiological Data

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Abstract

COVID-19 was declared a global health emergency in March 2020, and governments and public health agencies found themselves generating data at a scale no one had quite prepared for. The rapid spread of COVID-19 created an urgent demand for timely analysis of large-scale epidemiological data. Making sense of data quickly, and in forms that non-specialists could actually use became one of the defining challenges of the pandemic response. This paper describes the design and development of interconnected Tableau dashboards built to analyse India's COVID-19 epidemiological data from January to August 2020. Two publicly available datasets provided the foundation: a patient-level record covering 20,25,376 cases and a state-wise daily aggregation across all 36 states and Union Territories, supplemented by records for 267 registered testing laboratories. Raw data was cleaned and restructured using Python and the pandas library. Dates were standardised, age groups were binned, missing values were addressed, and status fields were pivoted into usable form. The resulting dashboards each tackle a distinct analytical question: spatio-temporal spread and cumulative case trends, transmission dynamics alongside demographic patterns, sub-national geographic and age-stratified rankings, and testing infrastructure distribution. The main findings are clear enough: community transmission had taken over by mid-2020, the 30–40 age group accounted for the most detections, Tamil Nadu and Maharashtra carried the heaviest cumulative burden, and the geographic overlap between laboratory density and reported case counts points to systemic underreporting in certain states. Taken together, the study offers a reproducible analytics workflow and fills a gap that the existing literature has left open, the absence of a multi-dimensional, cross-referenced visualisation platform for India's COVID-19 data.

Keywords

COVID-19 · Tableau · Data Visualisation · Dashboard · Epidemiology · Visual Analytics

Introduction

The SARS-CoV-2 virus has produced a pandemic that, by most measures, stands among the worst public health crises in modern history. Scientists around the world are still working on fundamental questions about the virus, including efforts to estimate true mortality across different populations [6]. Since its emergence in December 2019, the outbreak has generated an extraordinary volume of data. Governments, research institutions, and international agencies have all been producing records: infection counts, hospitalisation figures, death statistics, vaccination rollout data. In India specifically, the Ministry of Health and Family Welfare

(MoHFW), volunteers at covid19india.org and the Indian Council of Medical Research (ICMR) together maintained detailed patient-level records and state-disaggregated datasets throughout the crisis. With the pandemic disrupting nearly every domain of daily life, analytics tools have become important aids for scientists, policymakers, and health systems trying to understand what they are dealing with. Tableau fits naturally into this context. It can connect to a wide range of data sources with minimal friction and, crucially, allows users to build interactive dashboards without extensive programming expertise, a point Ko and Chang [4] specifically highlight as one of its strengths in healthcare settings. This paper builds on that potential. The goal was to develop a cohesive, integrated Tableau dashboard that draws on multiple COVID-19 datasets for India and to document the process and results so that stakeholders can grasp them.

Related Work

The literature about COVID-19 visualisation is quite extensive and includes different approaches such as epidemiological modelling, real-time dashboards, and visual analytics platforms; some of the contributions to this area are highly pertinent. N Akhtar et al. [1] offer an impressive description of Tableau features, including data blending, real-time reporting, and geospatial mapping. Another important contribution by Dong et al. [2] gives an in-depth overview of the COVID-19 dashboard created by the Johns Hopkins University; this is perhaps the most popular real-time tracker of the pandemic. The study demonstrates the importance of a scalable data pipeline infrastructure that allows handling inputs from multiple sources as well as adjusting to evolving needs, ranging from reporting cases to recording vaccinations. Moreover, the need for standardised data across jurisdictions was mentioned.

In relation to regional variations, Ghouzali et al. [3] find that comparison of regional growth rates and case-fatality rates makes it possible to discover local dynamics that cannot be found using aggregated data at the national level. In the same spirit, I. Ko and H. Chang [4] note that Tableau is an especially useful tool in healthcare because of its ability to combine diverse databases into interactive and informative dashboards that require no special programming knowledge. In this regard, S. Martha [5] shows how to use Tableau for generating localised risk assessments by analysing infections and demographic data to derive a score that would make it easier for clinicians to interact with patients.

Concerning the specificities of the Tableau application in India, Mitra et al. [6] offer an example of developing visualisations that incorporate epidemiological characteristics at the local level; however, their contribution is more concerned with epidemiological models than visual analytics, which might become especially important if the task involves exploring large data sets. Data accuracy issues remain another key problem when studying the COVID-19 situation in India. According to S. Natarajan and P. Subramanian [7], there are substantial discrepancies between officially recorded data and estimated true numbers. Therefore, any visualisation based on official data is inevitably restricted in scope. As an example of applying Tableau for Indian datasets, S. Vasundhara [8] reports creating informative graphics for key epidemiological variables.

Thus, there is sufficient literature proving the usefulness of Tableau for public health visualisation purposes. Still, a comprehensive dashboard covering Indian data related to demographic factors, time trends, geographical distribution, and testing infrastructure has yet to be created.

Research Gaps

Progress in the field is real, but important gaps remain. The most obvious one is the persistent separation between demographic analysis, i.e. people-centric analysis keeping individuals in the centre, and state-level trend analysis, i.e. high-level numeric analysis for a wider frame. Some studies have examined age distributions, and others have tracked region-wise case counts [3], but the two strands rarely appear together in a single visual analytics tool. Most published dashboards concentrate on headline aggregates: total confirmed cases, deaths, and recoveries. Age distribution, gender composition, and regional population characteristics tend to get ignored or treated in isolation. That's a missed opportunity, because demographic breakdowns by state could reveal which populations are most at risk and whether there are meaningful connections between demographic structure and how the virus has spread. Equally absent is any serious integration of testing infrastructure data with demographic and epidemiological findings. The number of confirmed cases in any region is not simply a function of how many people are infected but also depends on how many are tested. This seems like an obvious point, yet no study in the reviewed literature attempts to incorporate testing capacity into a unified visualisation framework.

A second problem is methodological transparency. The process of taking raw, inconsistent public health data drawn from official portals, crowdsourced repositories, and other sources and turning it into something Tableau can actually work with is neither trivial nor automatic. Cleaning, normalising, handling missing values, merging datasets from different platforms: these steps matter enormously, but existing studies rarely describe them in enough detail to allow replication. The result is that even where Tableau has been used effectively, the analysis remains a black box. Others cannot build on it without redoing all the preparatory work from scratch.

Third, and perhaps most consequential for interpretation: no published visualisation of India's COVID-19 experience takes testing infrastructure seriously as a variable. Reporting confirmed cases per million people is standard practice, but that figure is meaningless without knowing how many people were actually tested. States with limited laboratory capacity will systematically undercount infections, not because fewer people are sick but because fewer cases are found. No existing study maps testing infrastructure or examines its relationship to reported case trends, which means the geographic patterns visible in nearly every published dashboard may reflect testing capacity as much as actual disease spread.

Design and Development of COVID-19 Visualisation Dashboards

To fill in these gaps, a two-step approach involving data preprocessing and interactive visual analytics was employed. Specifically, the initial stage involved the acquisition, cleaning, transformation, and preparation of datasets. Raw COVID-19 data obtained from multiple publicly accessible resources were incorporated into an analysis-friendly framework. Missing and inconsistent data entries were detected and corrected using proper data-cleaning approaches. The duplicates were eliminated, data types were standardised, and date columns, categorical variables, and numerical indicators were transformed into a form suitable for further analysis. Additionally, several feature engineering and categorisation operations were performed in order to organise records by certain parameters such as the region, age group, gender, type of transmission, status of infection, etc. Overall, the pre-processing stage was aimed at preparing the dataset for efficient visualisation and integration with the interactive dashboards.

In the next step, the prepared dataset was imported into Tableau Desktop Public Edition to perform visual analysis and dashboarding operations. Several interactive dashboards were developed, each providing insights into different aspects of COVID-19, such as time dynamics, demographics of the population affected by the virus, geographic distribution, infection transmission, and other factors. Charts, maps, filters, and various types of statistics were utilised to increase interpretability and enable comparative analysis of different variables included in the database. Furthermore, Tableau Dashboard Actions were employed to make dashboards interconnected and provide users with an opportunity for dynamic interactions with the visual content. Through Tableau Actions, any selection made in a certain visual element would automatically apply to connected visual elements.

Datasets Used

Table 1.1 gives a quick view of the datasets used for this study and analysis. Both primary datasets came from publicly accessible collections on Kaggle. Dataset 1 is a patient-level record sourced from MoHFW and related repositories, covering variables including Patient Number, State/UT, District, City, Age, Gender, Date of Detection, Transmission Type, and Status. Dataset 2 provides state-wise daily figures for confirmed cases, recoveries, and deaths across all 36 states and Union Territories. Dataset 3 includes records for 267 registered testing laboratories across the country.

Table 1.1. Summary of Datasets Used

Dataset	Granularity	Primary Use
COVID-19 in India (Patient-level)	Patient record	Demographics, transmission type, and detection peaks
COVID-19 India (State-wise daily)	State/Daily Records	Trend lines, geographic map, cured vs. deaths
Testing Labs	Lab record	Lab locations, types, and state-wise counts

Exploratory Data Analysis

Before building any dashboards, each dataset was profiled to understand its structure and quality. Table 1.2 summarises the key figures that emerged from this initial review.

Table 1.2. Key Dataset Statistics

Metric	Value
Total patient records	20,25,376
Date range	January 2020 – August 2020
States / UTs covered	36
Total confirmed cases	20,25,376
Total recovered	13,70,487
Total active cases	6,05,900
Total deaths	41,310
Untraced cases	7,679
Testing labs	267
Government laboratories	182
Private laboratories	82
Collection sites	3

The patient-level dataset had a noticeable proportion of null values in the District and City fields, especially for records from smaller states and Union Territories. Age data was largely complete but missing for a subset of records; those entries were excluded from the age distribution view but kept in all other visualisations to avoid unnecessarily reducing the overall case count.

Data Preprocessing

The preprocessing step entailed cleaning and transforming data for better accuracy and consistency in order to create an effective dashboard in Tableau. Raw data on cases of COVID-19 were cleaned and structured using Python programming and Pandas through techniques such as standardising dates, addressing missing data, restructuring categorical data, and data validation. Age intervals were also created to facilitate age-related analyses. Geographic data that did not include all components was preserved and marked appropriately without altering any data within the set. The patient status field was also rearranged for comparison purposes among states and categories. Figure 1 describes the flow of data and processes related to it appropriately.

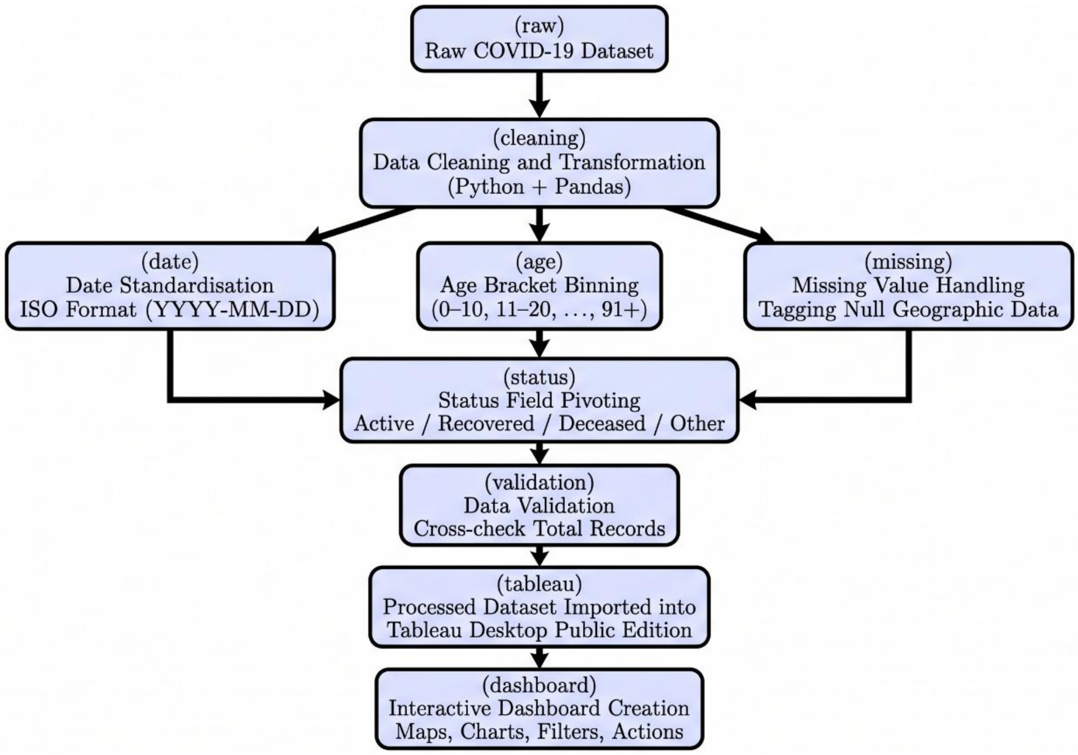

Figure 1: Dataflow for the Analysis.

Date Standardisation

All date variables were converted to ISO format (YYYY-MM-DD) using pandas. Month and year were then extracted as separate calculated fields inside Tableau to support the temporal aggregation needed for the detection peak bubble chart.

Age Bracket Binning

Age was stored as a continuous integer in the raw dataset. A new binned field was created in Tableau, grouping values into ten-year intervals (0–10, 11–20, …, 91+) for the age distribution histogram. Records with missing or implausible age values were excluded from that particular view.

Missing Value Handling

Records lacking district or city information were tagged as ‘Null’ in the geographic ranking charts but were retained in all other views rather than dropped from the dataset entirely.

Status Field Pivoting

The Patient Status field, covering active, recovered, deceased, and migrated/other, was pivoted to produce a table breaking down patient counts by state and status. The resulting totals were cross-checked against the full patient count (SUM(CNTD(Patient Number)) = 20,25,376) to confirm consistency.

Tools and Technologies

The idea of tools that have been utilised to perform several functions for the preprocessing and development of dashboards is discussed in Table 1.3. NumPy and Pandas are the names of Python libraries that help in handling and preprocessing data. Tableau Desktop refers to software used for analysing and designing dashboards. Tableau Public refers to the web-based tool where visualisation can be publicly accessed. Mapbox refers to the connector software used in Tableau Desktop for geographical mapping of raw data.

Table 1.3. Summary of Tools Used

Tool / Technology	Version / Edition	Purpose
Python 3: Pandas, NumPy	Python 3.x	Date standardisation, null handling, field pivoting
Tableau Desktop Public Edition	Public Edition	Dashboard authoring, calculated fields, cross-filter actions
Tableau Public	Web Platform	Dashboard publication and online sharing
Map box (via Tableau)	Integrated	Geographic rendering for patient case and lab maps

Dashboard Architecture

The final product comprises three primary dashboards and one supplementary dashboard, each tackling a distinct analytical question while sharing a consistent visual design. All were built in Tableau Desktop Public Edition at a minimum size of 420 × 560 pixels. They are connected through Tableau's dashboard actions, selecting a state or region in one view automatically filters the others. The dashboards are:

1. Spread and Trend Analysis
2. Transmitting Dynamics and Demographic Analysis
3. Spatial Ranking and Age-Based Analysis
4. Laboratory Facilities Distribution

Dashboard Analysis and Interpretation

Spatio-Temporal Spread and Cumulative Trend Analysis Dashboard

Dashboard shown in Figure 2 addresses the spatial and temporal dimensions of the outbreak together. It brings together visualisations: a geographic spread map with a month-of-detection filter, a multi-line chart tracking cumulative case trends by state, a dual-series chart plotting daily newly detected cases against daily recoveries, and a pie chart comparing total recoveries to total deaths.

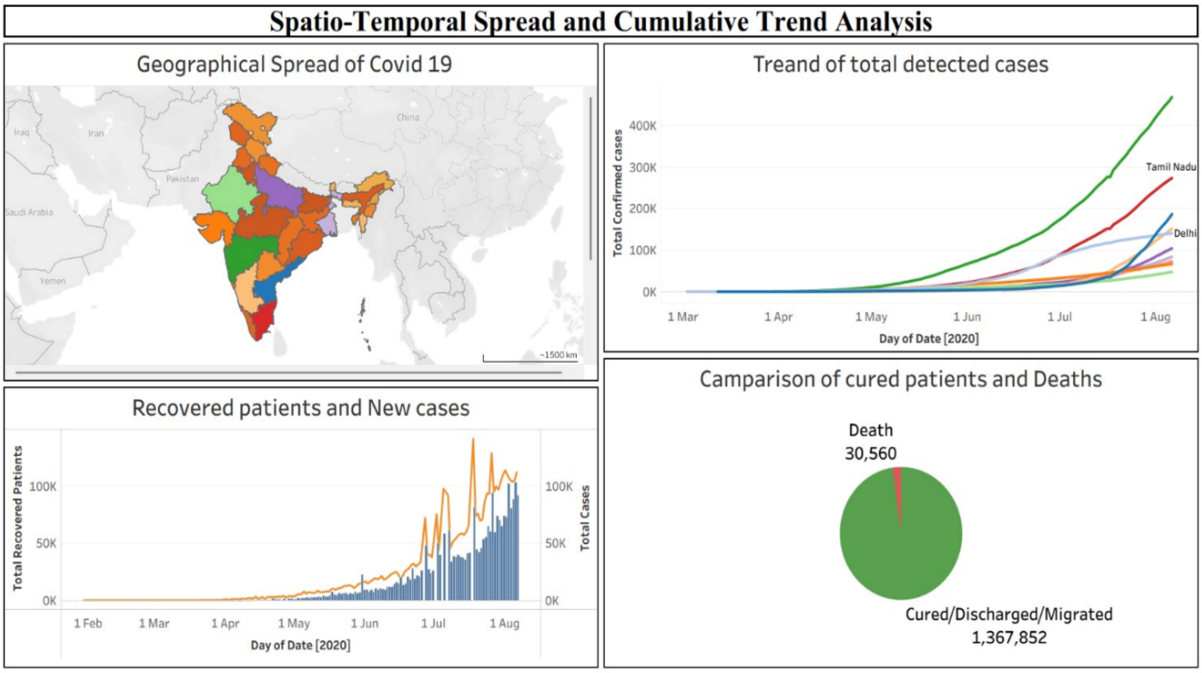

Figure 2: Spatio-Temporal Spread and Cumulative Trend Analysis Dashboard

Geographic Spread Map

Figure 3 shows the geographical spread of COVID-19 across nations. Filtered by month from January through August 2020, the map traces the outbreak's geographic progression. Cases appeared first as a small cluster in Kerala in January and February, then expanded outward month by month. By May–June, all major peninsular states had recorded cases. By July–August, the spread had reached almost the entire subcontinent, with the heaviest concentration of markers in Maharashtra, Tamil Nadu, Delhi, and Uttar Pradesh. The image table below captures this month-by-month geographic progression across states.

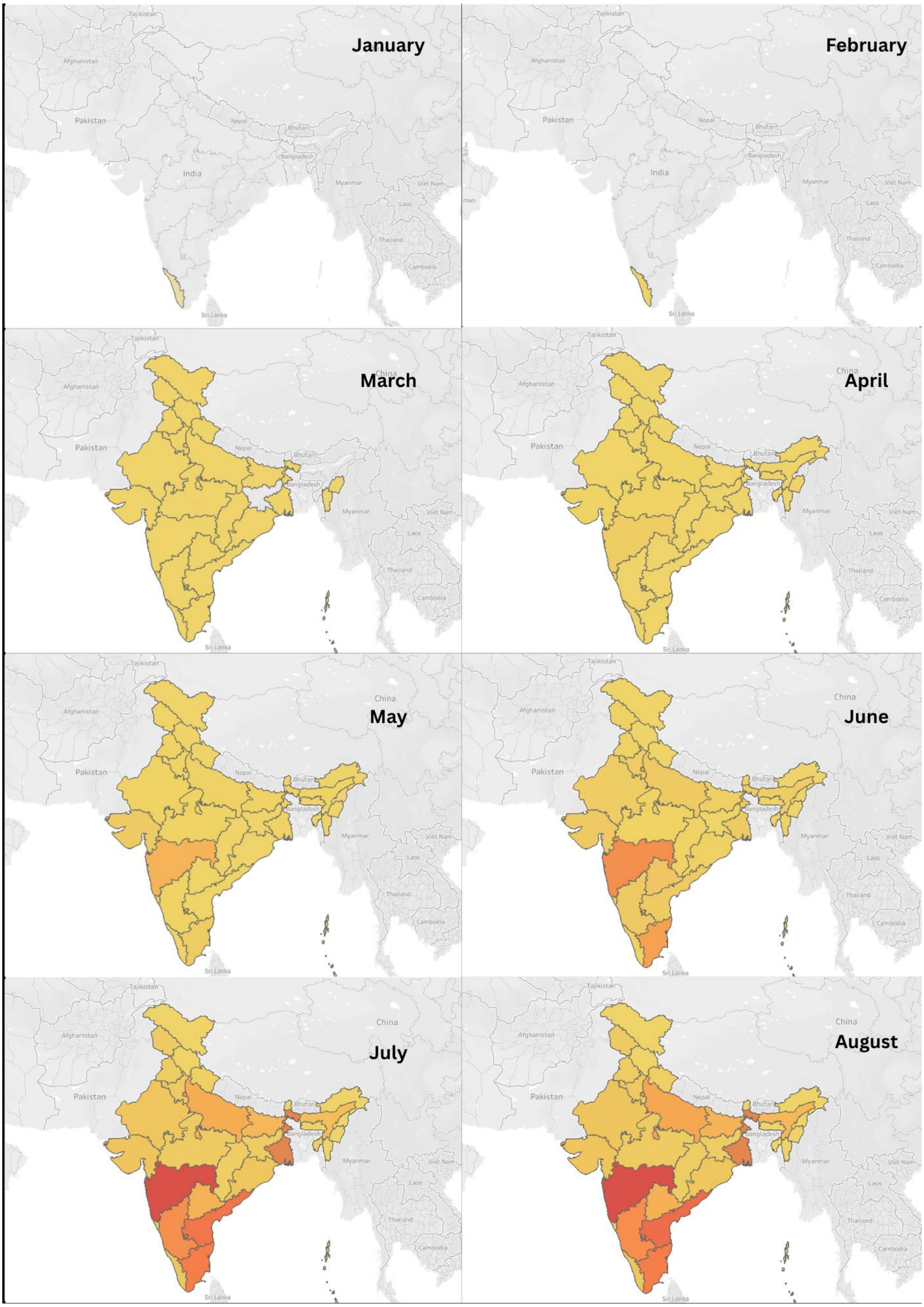

Figure 3: Progression of COVID-19 Spread Across States in 2020

Cumulative Case Trend (Multi-line Chart)

Figure 4 follows the state-wise trend of total detected cases of COVID-19. Each state appears as a separate line tracking its cumulative case count over the study period. From May onward, the lines begin to pull apart noticeably. A handful of high-burden states, Maharashtra, Tamil Nadu, and Delhi, most prominently trace a steep upward curve approaching 4,00,000 cases, while the majority of states remain below 1,00,000 through the end of August. As we will see in further analysis, these are the states with a high density of testing facilities.

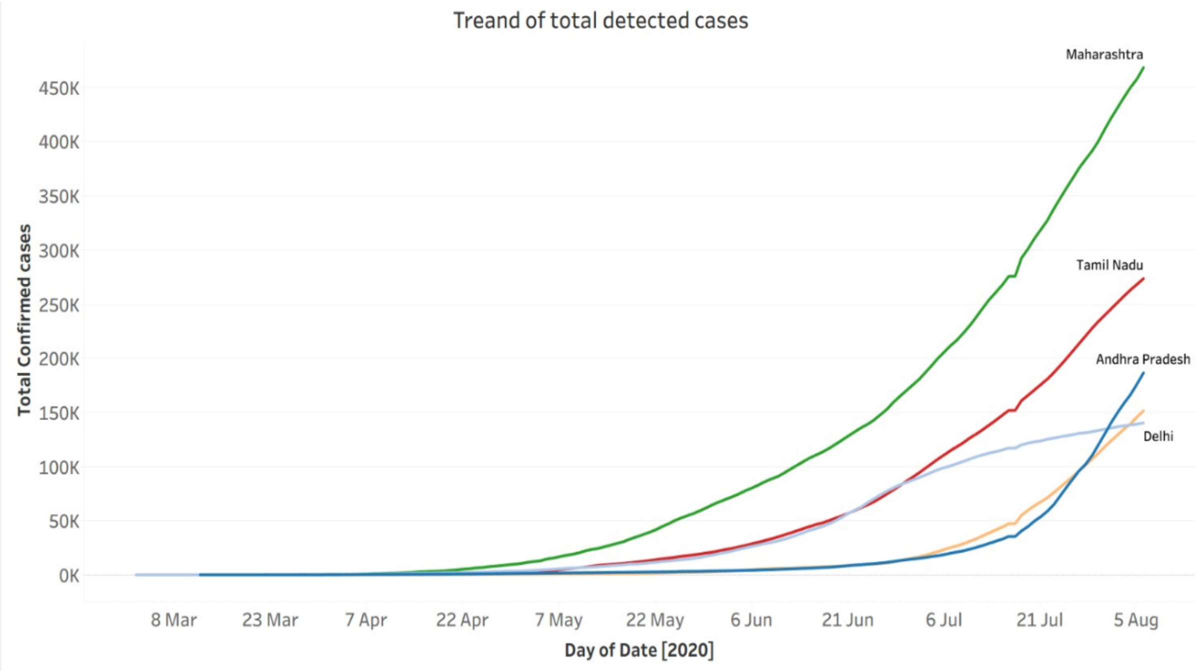

Figure 4: State wise Case Detection Trend.

Daily Cases vs. Daily Recoveries (Dual-Series Chart)

Figure 5 explores the comparison of COVID-19 detection and recovery trends. Two lines track the full study period: new daily cases and new daily recoveries. Through April and May, the two move in rough tandem. From June, the gap widens as new cases outpace recoveries, a divergence that peaks in July. Then the lines begin converging again through August, reflecting an improving recovery rate in the second half of the period.

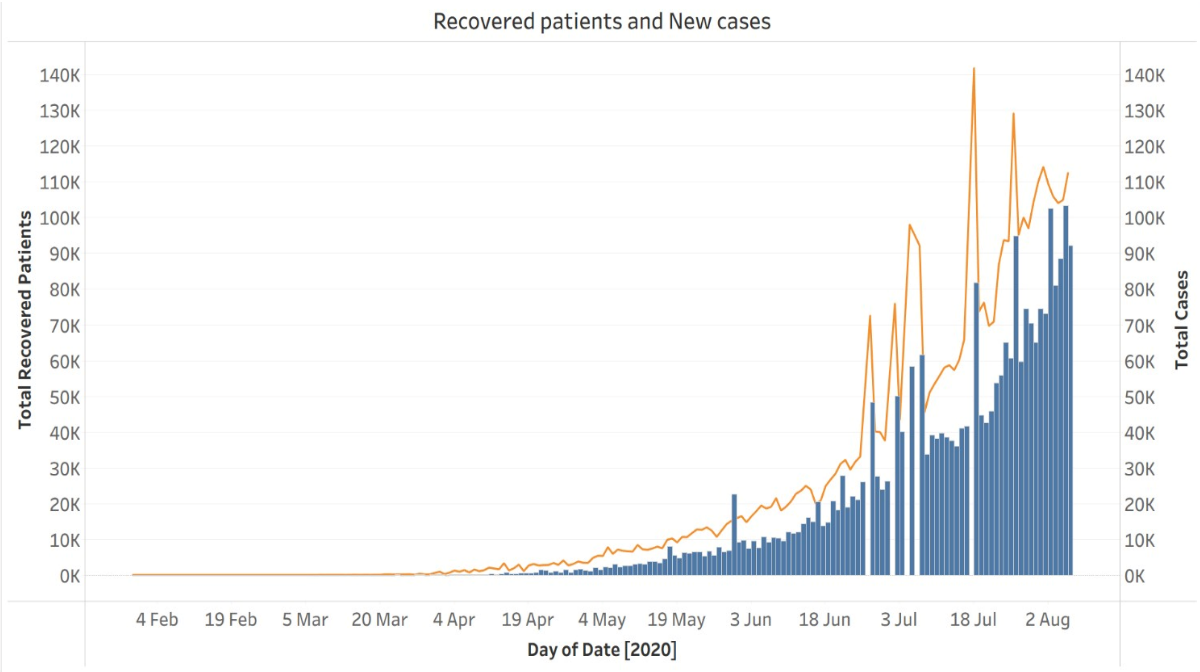

Figure 5: Covid-19 Detection and Recovery Trend.

Cured vs. Deaths (Pie Chart)

Figure 6 depicts the outcome of detected cases. The total cases recorded during the study period, 13,70,487, resulted in recovery against 41,310 deaths, a case fatality ratio of 2.93% for the initial 8-month period of spread. Filtered outputs also indicate a high fatality rate in states with a high detection rate, indicating possible under-reporting of fatalities.

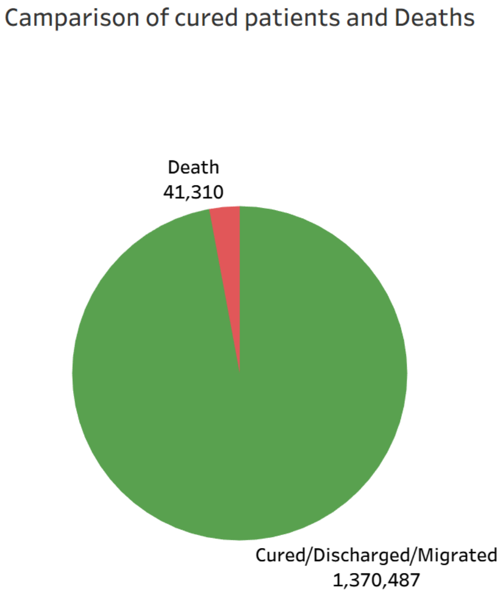

Figure 6: Covid-19 outcomes.

India's epidemic followed a recognisable diffusion pattern: entry through coastal states, then gradual spread inland along population corridors. The cumulative burden has remained highly concentrated, and a small number of heavily urbanised states account for a disproportionate share of total cases. The widening gap between new cases and recoveries during June–July reflects a period of genuine system strain, one that Ghouzali et al. [3] note as a recurring feature of regional outbreak dynamics before stabilisation sets in.

Transmission Dynamics, Detection Peaks, and Demographic Profiling

Where the first dashboard focused on aggregate trends, Dashboard shown in Figure 7 zooms in on how the virus actually moved when detections peaked, what drove transmission, and who was most often diagnosed. Visualisations make up the view: a bubble chart of monthly detection peaks, a pie chart of transmission type, a state-wise case status distribution, and a pie chart for gender breakdown.

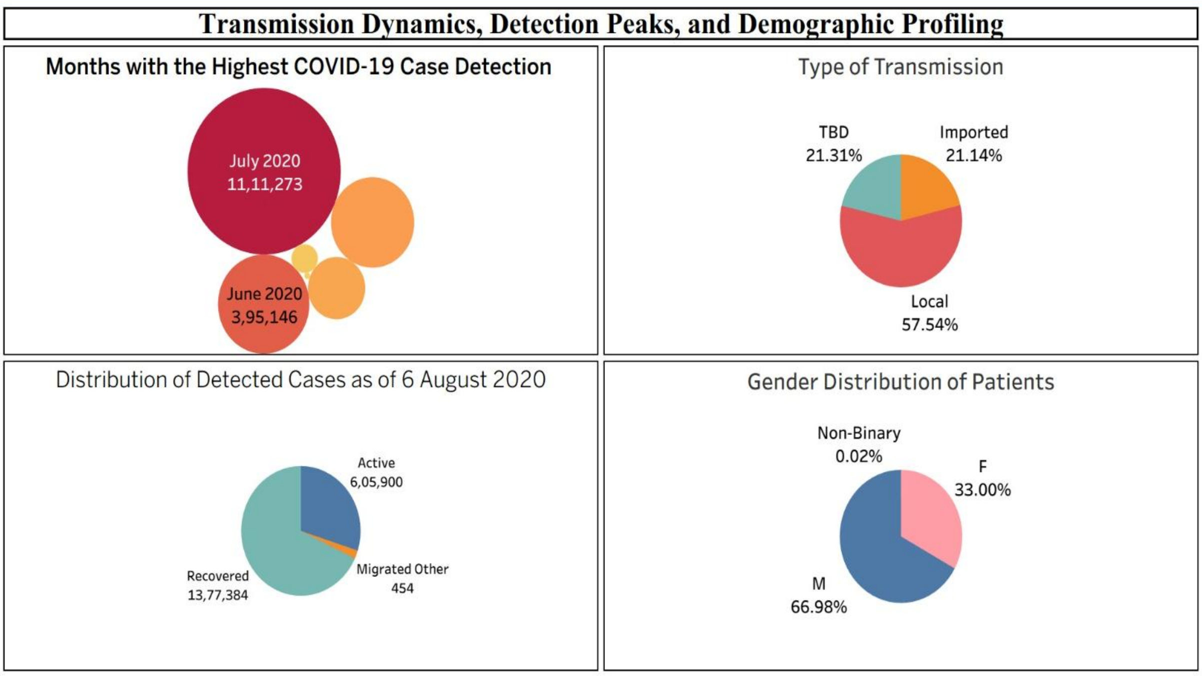

Figure 7: Transmission Dynamics, Detection Peaks, and Demographic Profiling Dashboard

Monthly Detection Peaks (Bubble Chart)

Figure 8 indicates the top months by COVID-19 case detection. Each bubble represents one month, sized by detection volume. The data is sharply skewed: July 2020 alone recorded 11,11,273 detections, approximately 2.8 times the June total of 3,95,146, and greater than all pre-July months combined.

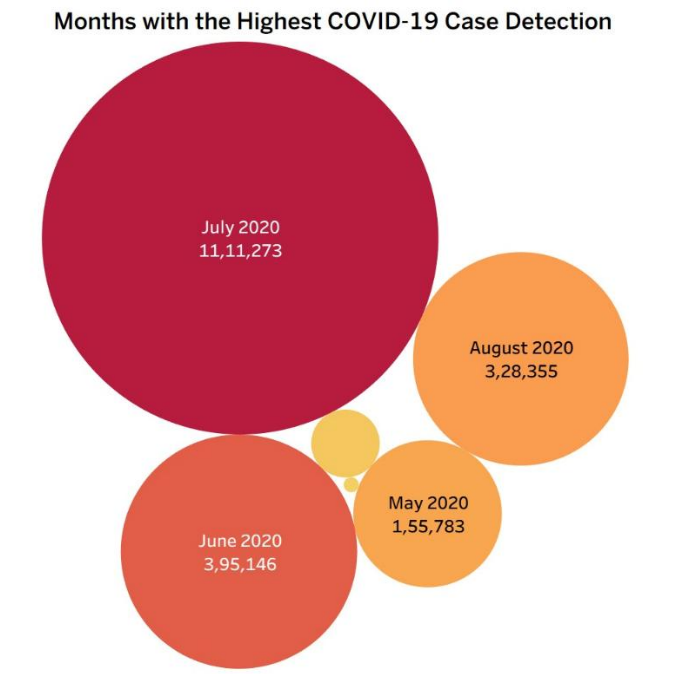

Figure 8: Highest Months by Covid-19 Detections.

Transmission Type (Pie Chart)

Figure 9 clearly concludes that the major mode of transmission in India is Local Transmission. The initial spread was through travellers arriving in Kerala from other countries. By mid-2020, local community transmission had become the dominant mode the imported cases and known-contact infections that characterised the early weeks had been overtaken entirely. For 21.13% of cases, the mode of transmission was not determined (To Be Determined), indicating weak monitoring and tracking infrastructure.

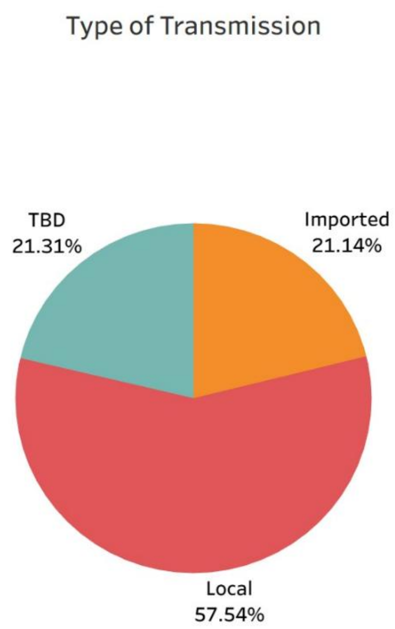

Figure 9: Highest mode of transmission of Covid-19.

Case Status Distribution

Figure 10 represents the status of patients as of August 2020, excluding deceased patients. Recoveries (13,70,487) make up the largest group. Active cases at the time of data compilation (6,05,900), deaths (41,310), and a small number of migrated or otherwise-classified cases (454) account for the remainder. The balance between these groups varies across states; several high-burden states show proportionally more active cases at the time.

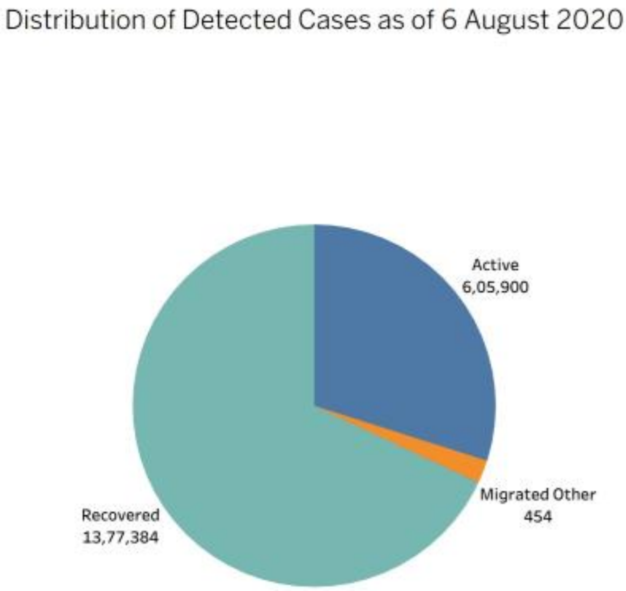

Figure 10: Status of Patients (As of August 2020).

Gender Distribution (Pie Chart)

Figure 11 states that male patients account for the majority of identified cases, a pattern consistent with findings from other countries [6]. The reasons are likely multiple: occupational exposure, differences in health-seeking behaviour, a high percentage of the workforce in essential services and possibly biological factors all play some role. A very low percentage of non-binary patients could be due to a lack of social factors related to identification.

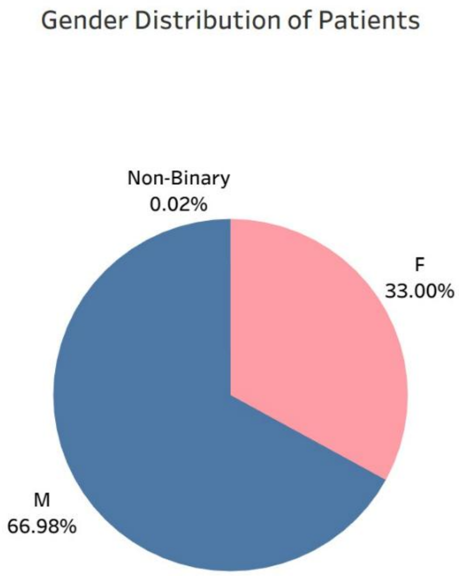

Figure 11: Covid-19 spread across genders.

The speed of the shift to community transmission effectively ended contact tracing as a viable containment strategy by mid-2020. The July surge is harder to interpret cleanly; it could reflect a real acceleration in cases, expanded testing capacity, accumulated reporting delays, or some combination of all three. The gender gap warrants further investigation, particularly with respect to occupational exposure patterns in urban areas.

Sub-National Geographic Ranking and Age-Stratified Distribution

Dashboard shown in Figure 12 aggregates the data geographically and demographically. It presents views: an age distribution histogram, a tree map ranking top cities, a bar chart for top districts, and a second tree map for state-level rankings. This dashboard highlights spread hotspots like cities, districts and states with the highest spread, possibly alerting authorities in time to step up reconsolidation efforts.

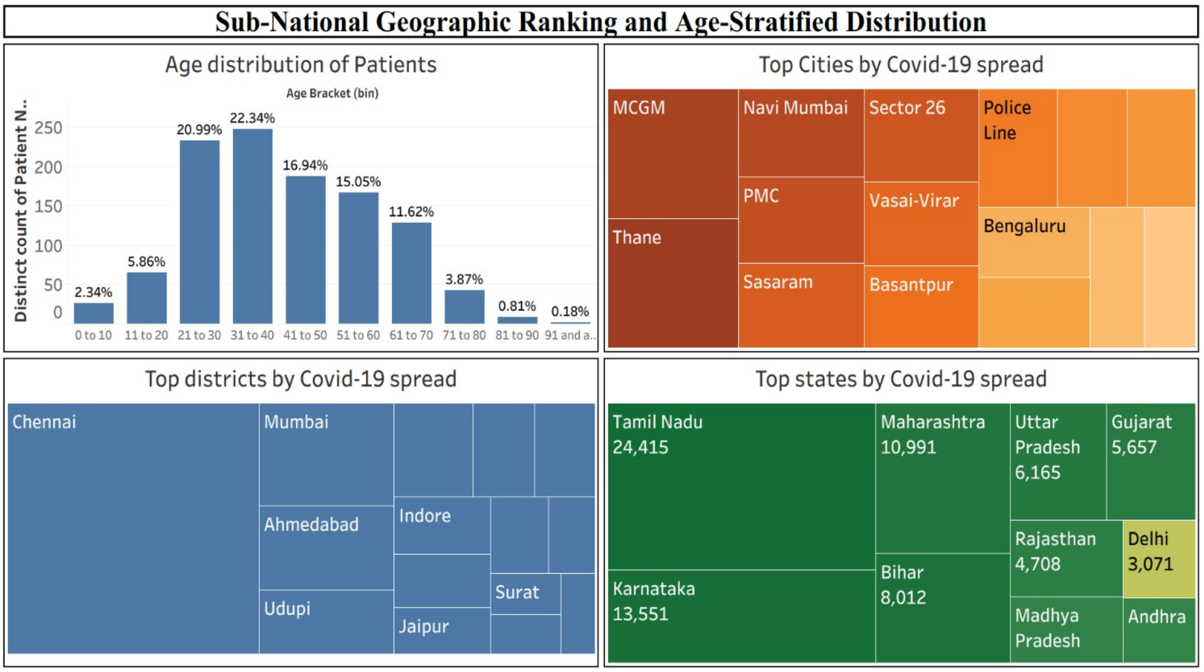

Figure 12: Sub-National Geographic Ranking and Age-Stratified Distribution Dashboard.

Age Distribution Histogram

Figure 13 shows the age distribution of the COVID-19 patients. Records are grouped into ten-year age bins. The distribution is unimodal, peaking sharply at the 31–40 bracket with the adjacent 20–30 and 40–50 groups also well represented. Detection rates are lowest at the extremes, the 0–10 and 70+ groups, which likely reflects lower incidence among children, and under detection of asymptomatic paediatric cases.

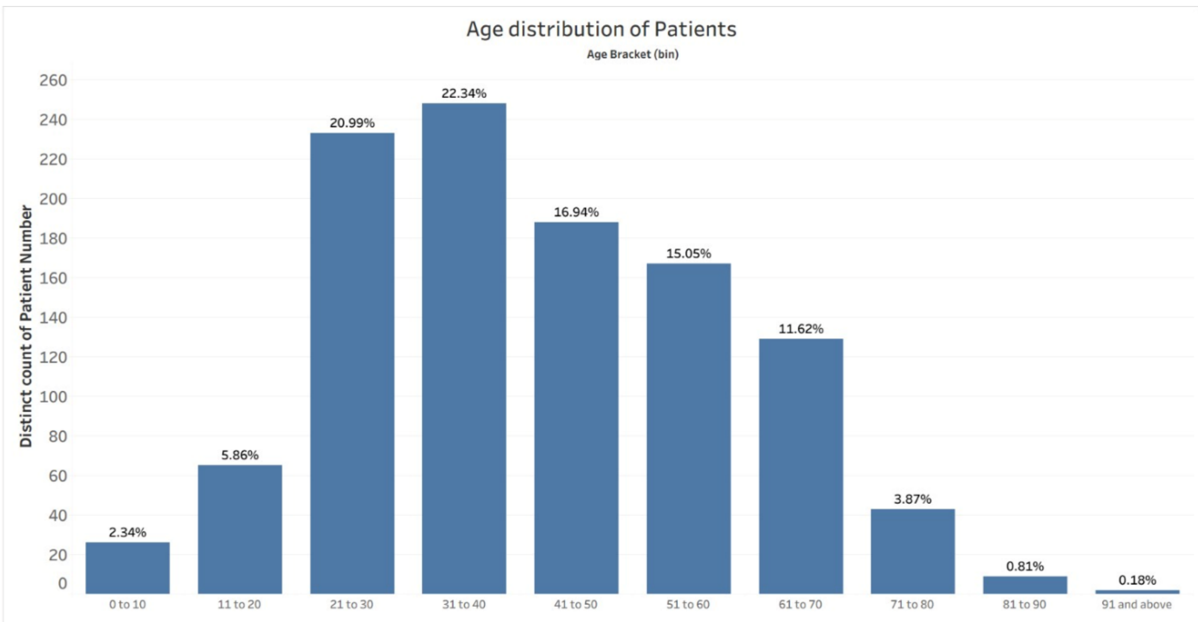

Figure 13: Covid-19 Age Demographics.

Cities with the highest spread (Tree Map)

Figure 14 alerts about cities with the highest spread. MCGM (Brihanmumbai Municipal Corporation) dominates the tree map, followed by Thane, Navi Mumbai and PMC (Pune Municipal Corporation). Maharashtra's municipal corporations collectively occupy much of the chart, reflecting both genuine case concentration in these dense urban areas and the fact that their relatively developed testing infrastructure ensures more cases are detected and recorded.

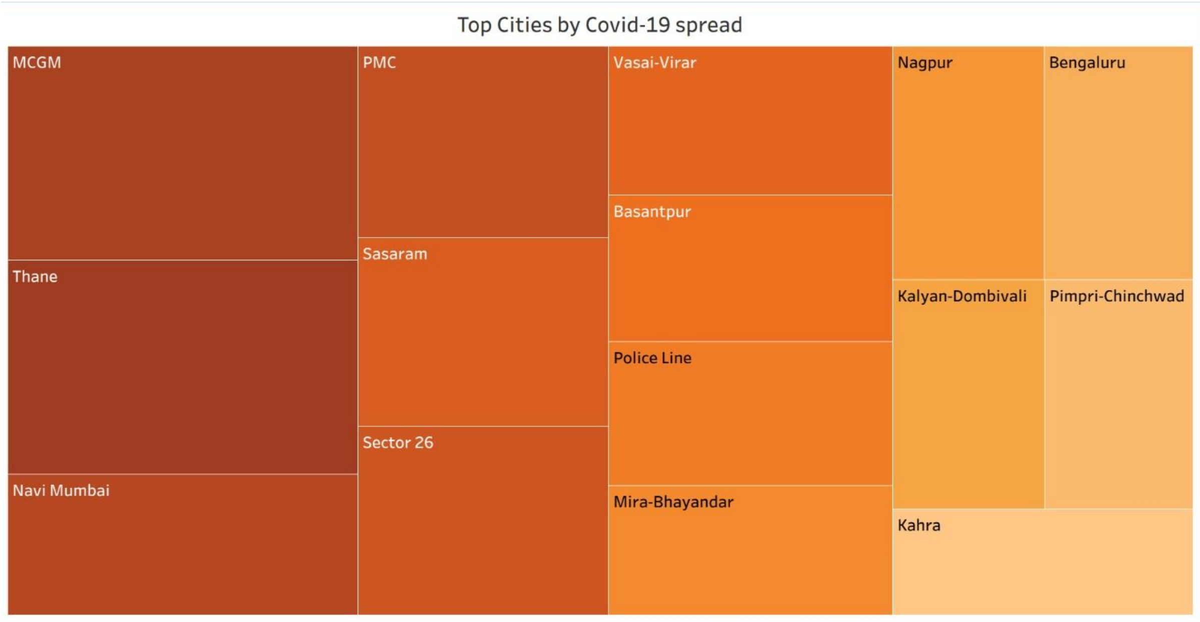

Figure 14: Top Cities by Covid-19 Case Detections.

Districts with the highest spread (Tree Map)

Chennai leads at the highest spread on the district level, followed by Mumbai and Ahmedabad. The ordering is not coincidental; all three rank among India's most well-equipped urban districts for laboratory infrastructure, as the testing dashboard 4 confirms. Figure 15 can help district administrations to take the required actions to limit the spread.

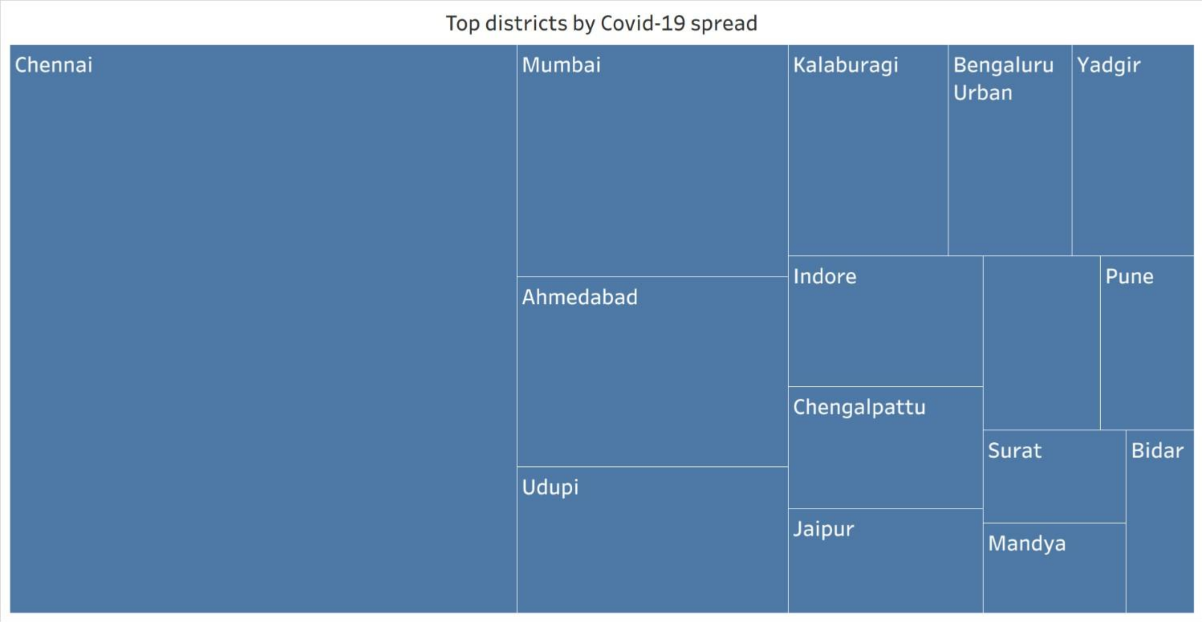

Figure 15: Top Districts by Covid-19 Case Detections.

States with the highest spread (Tree Map)

From Figure 16, we can see that at the state level, Tamil Nadu takes the top position, ahead of Maharashtra, Bihar, and Uttar Pradesh. While Maharashtra dominates the city-level rankings, Tamil Nadu's broader geographic distribution of cases spread across more districts gives it a higher aggregate state figure.

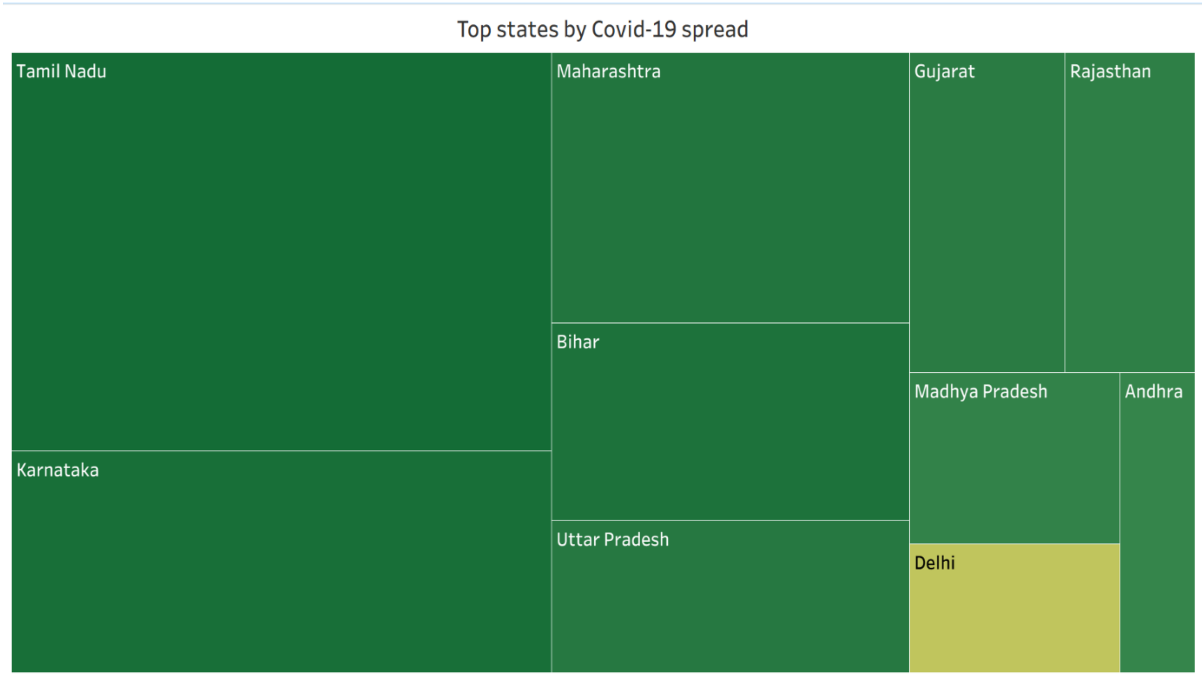

Figure 16: Top Stats by Covid-19 Case Detections.

India's outbreak was, at its core, an urban phenomenon concentrated in the working-age population. The 30–40 peak in the age distribution aligns with the demographic profile of

India’s urban workforce, pointing toward occupational exposure as a primary driver. The spatial correlation between testing capacity and case counts is strong enough to raise a serious interpretive caution: the data tells us where cases were found, not necessarily where the disease was most prevalent.

Testing Infrastructure, Distribution, and Laboratory Capacity Mapping

Testing capacity is the analytical foundation that makes any case count interpretable. Dashboard in Figure 17 maps that infrastructure directly, using three views: a point map of all testing laboratory locations, a stacked bar chart showing lab counts by state, and a national pie chart breaking down lab types.

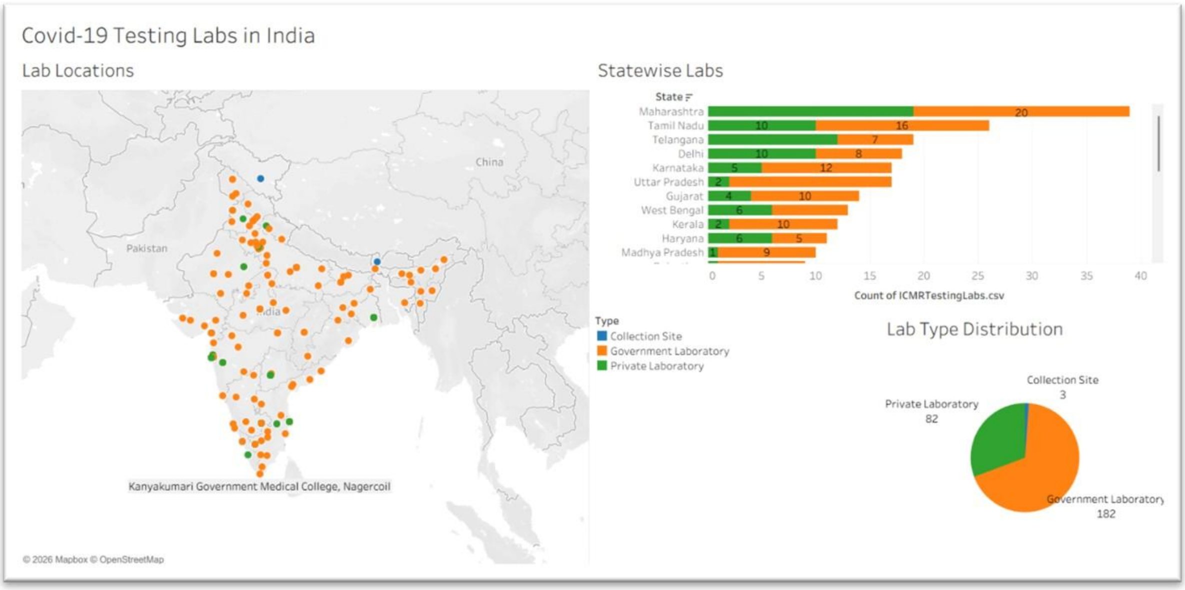

Figure 17: Testing Infrastructure Distribution and Laboratory Capacity Mapping Dashboard.

Geographic Point Map of Testing Laboratories

Each of the 267 registered laboratories appears as a point on the map, colour-coded by type: government, private, or collection site. The concentration is striking. Maharashtra, Tamil Nadu, Karnataka, and Kerala together hold a disproportionately large share of laboratories relative to both their geographic size and population. By contrast, large and highly populated inland states like Uttar Pradesh, Madhya Pradesh, and Rajasthan have far fewer labs per square kilometre. This directly drives testing capacity and, in turn, lowers case detection in these states. There is a serious possibility that a large number of COVID-19 cases went under-reported in some of the regions.

Number of Labs by State – Stacked Bar Chart

Maharashtra leads with 40 labs, followed by Tamil Nadu (26), Delhi (18), and Telangana (15). Government laboratories dominate in most states. Economically stronger states, particularly Maharashtra and Tamil Nadu, also show a higher share of private facilities. Many Union Territories and smaller states have only one or two laboratories, which means their testing numbers are heavily shaped by sample referral practices rather than local testing capacity.

National Type Distribution of Labs – Pie Chart

Of the 267 facilities, government laboratories account for 68% (182 labs), private labs for 31% (82 labs), and collection sites for just 1% (3 labs). The heavy reliance on government infrastructure means that the national testing scale was essentially determined by how quickly government labs could expand a structural constraint, with direct implications for where and how quickly cases were identified.

The overlap between high lab density and high reported case counts, concentrated in southern and western coastal states, introduces a systematic reporting bias that shapes the geographic patterns visible in other dashboards [8]. States with fewer laboratories report fewer cases, but this reflects detection capacity, not disease prevalence. The excess mortality evidence reinforces this point: S Natarajan and P Subramanian [7] document substantial gaps between official figures and estimated true deaths, and despite methodological differences across studies, the researchers converge on the same conclusion: underreporting of both cases and deaths was widespread.

Summary of Key Findings

Table 1.4 summarises observations derived from dashboard analysis in short.

Table 1.4. Summary of Key Findings

Finding	**Detail**
Temporal surge	July 2020 peak: 11,11,273 detections (~2.8× June 2020 total of 3,95,146)
Geographic concentration (state)	Tamil Nadu, Maharashtra, Bihar, Uttar Pradesh
Geographic concentration (city/district)	MCGM, Thane, PMC; Chennai, Mumbai, Ahmedabad
Demographic peak	30–40 age bracket; male majority
Dominant transmission mode	Local community transmission by mid-2020
Fatality ratio	~2.93% in aggregated dataset (41,310 deaths / 20,25,376 cases)
Testing infrastructure	Government labs 68%, private labs 31%.
Detection bias	Lab density co-located with high case-count states, undercounting likely in certain states [3]

Conclusion

This study developed interconnected Tableau dashboards covering India's COVID-19 outbreak from January to August 2020, drawing on patient-level demographic records, state-aggregated daily statistics, and laboratory infrastructure data. The combination of Python-based preprocessing with Tableau's visualisation capabilities demonstrates a repeatable analytics workflow that can function in resource-constrained settings, a point N Akhtar et al. [1] make broadly about Tableau's accessibility for public health applications.

The analysis paints a consistent picture. By the second half of 2020, community transmission had become the dominant mode of spread in India, concentrated in urban areas and most prevalent among people aged 30–40. The aggregate case fatality ratio sits at 2.93%, but that figure needs to be read alongside the testing infrastructure data: interior states with sparse laboratory coverage almost certainly have higher true mortality than official records show, a pattern consistent with findings on excess mortality [8].

The study directly addresses a gap that the reviewed literature left open: the absence of a cross-referenced, multi-dimensional Tableau dashboard for India that brought demographics, temporal trends, geographic patterns, and testing infrastructure together in a single platform. Future work could extend this foundation by incorporating vaccination rollout data or additional contextual variables, such as regional oxygen production capacity factors that became increasingly relevant as the pandemic evolved beyond 2020.

Competing Interests

The authors have no conflicts of interest to declare that are relevant to the content of this paper.

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